EXECUTIVE CAPSTONE PRESENTATION • SALES ANALYTICS

Business Performance Analytics

Turning sales and customer data into clear, actionable business insights

COMBINED REVENUE	TOP CATEGORY	TOP CUSTOMER
■23,600	Electronics	■4,000
Electronics + Accessories	■15,800 revenue	Karan • lifetime revenue query

Prepared by: Rishika Pattanayak | Tools: Python • Pandas • MySQL • Tableau

Methodology

A repeatable workflow from raw records to an executive-ready view

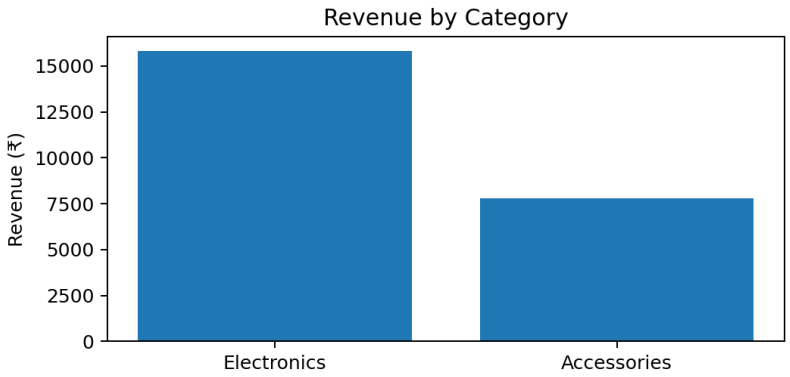
01 DATA INPUT	Sales, customer, product and inventory tables were used as the analysis inputs.
02 DATA PREPARATION	Python and Pandas supported inspection, cleaning and preparation of the data.
03 SQL ANALYSIS	MySQL queries used joins, aggregations and customer / revenue summaries.
04 KPI & TREND REVIEW	Revenue totals, category contribution, top customer and monthly revenue were calculated.
05 VISUAL REPORTING	Dashboard views communicate KPI cards, category comparison and monthly trend.

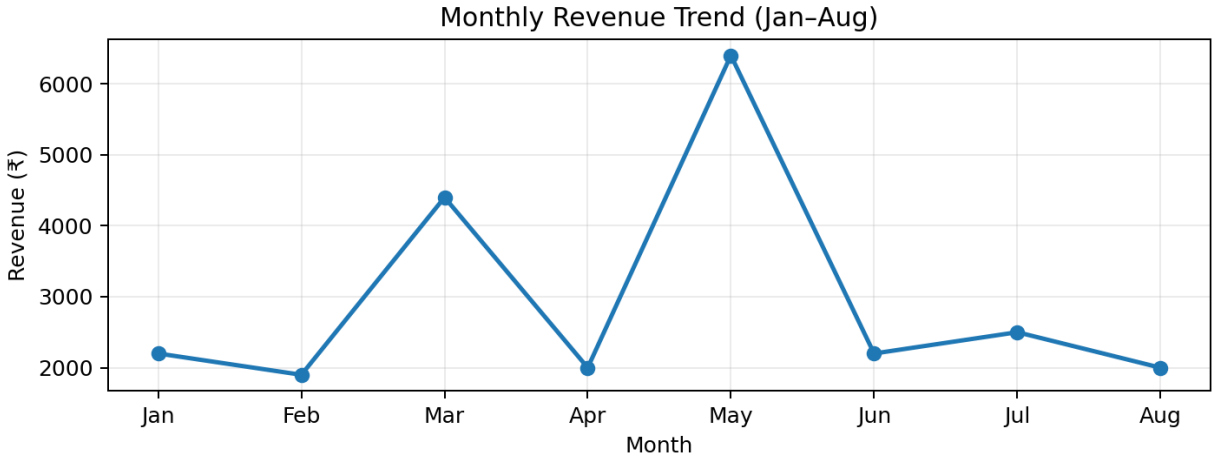
Workflow: collect → inspect & clean → query → calculate KPIs → visualise → interpret.

Key Findings

The query outputs show category concentration and a clear monthly revenue peak

COMBINED REVENUE	TOP CATEGORY	TOP CUSTOMER
■23,600	Electronics	■4,000
Electronics + Accessories	■15,800 revenue	Karan • lifetime revenue query





May is the strongest month in the Jan–Aug series (■6,400); Electronics contributes more than Accessories.

Measurement & Statistical Significance

Separate descriptive KPIs from claims that require additional data or formal testing

MAU Monthly unique buyers Count distinct customers in each month.	**CLV PROXY** Historical revenue Sum of recorded sales per customer; not a predictive lifetime-value model.	**CAC** Not calculable yet Marketing / acquisition spend is not available.

STATISTICAL SIGNIFICANCE: NOT ESTABLISHED

The completed work is descriptive analysis (counts, sums, rankings and trends). No formal hypothesis test or p-value was calculated for this capstone dataset. Therefore, we can describe the observed patterns, but we cannot claim that the differences are statistically significant or prove that one factor caused another.

Next measurement step: collect marketing spend and enough comparable time periods, then define and run an appropriate statistical test.

Strategic Recommendations

Three practical actions based on the patterns visible in the current analysis

01 PRIORITISE STRONG CATEGORIES	Review inventory and merchandising for higher-revenue categories, especially Electronics. Use stock availability and margin alongside revenue before making purchasing decisions.
02 STRENGTHEN CUSTOMER RETENTION	Create repeat-purchase campaigns for customers with previous purchase activity. Track monthly unique buyers and repeat-purchase rate to evaluate results.
03 MONITOR MONTHLY PERFORMANCE	Review monthly revenue and active customers regularly. Investigate spikes such as May and check whether they repeat before changing plans.

Conclusion: build decisions on consistently tracked KPIs, and validate patterns before treating them as business rules.